

How We Score Customers

RFM Methodology · Dunnhumby: The Complete Journey

R — Recency

Days since last purchase
Score 5 = shopped yesterday
Score 1 = last seen 658 days ago
Reference day = 712

F — Frequency

Distinct shopping trips
Score 5 = 1,300 trips
Score 1 = 1 trip
Method: `pd.qcut` quintiles

M — Monetary

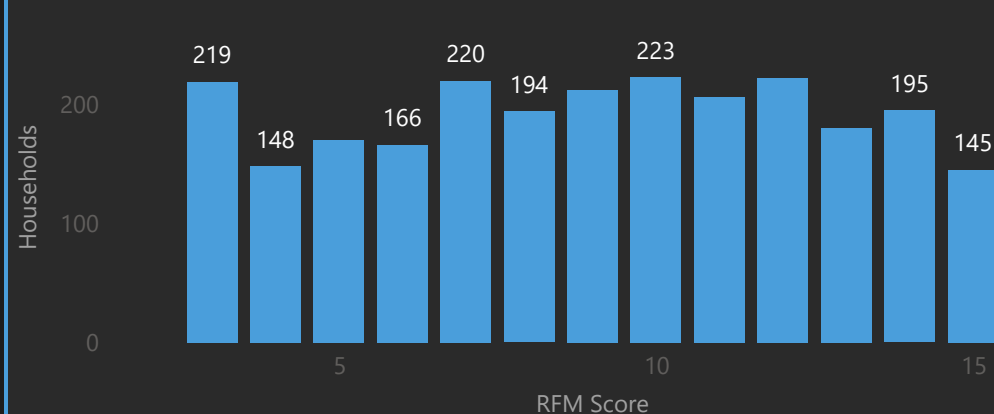
Total spend over 2 years
Score 5 = \$38,319
Score 1 = \$8
Excludes \$0 promo items

RFM Score = $R + F + M$
Range: 3 (worst) → 15 (best)

8 Customer Segments — Rule-Based Classification

Segment	Count of household_key	Average of RFM_Score
Champions	509	13.84
Loyal	330	11.53
Potential Loyalists	306	9.81
At Risk	565	8.36
New Customers	30	7.33
Low Value	62	5.45
One-time Buyers	198	5.07
Lost	500	4.73

RFM Score Distribution



Dataset Highlight

2500

Households

711

Transaction window(Days)

0

Uncategorised households

\$2,158

Median 2-Yr Spend

1300

Most Trips by a household

658

Longest Absence (days)

Scoring method: `pd.qcut()` quintile binning
Equal population per bucket (~500 households each)
Recency inverted: lower days = higher score

RFM Customer Segmentation — Executive Summary

Dunnhumby · 2,500 Households · Jan 2017 – Dec 2018 · Retail Grocery · USA

2,500

Total Households

\$8.06M

Total Revenue (2 Yrs)

42.6%

Churn Rate (At Risk + Lost)

\$2.14M

Churn Revenue

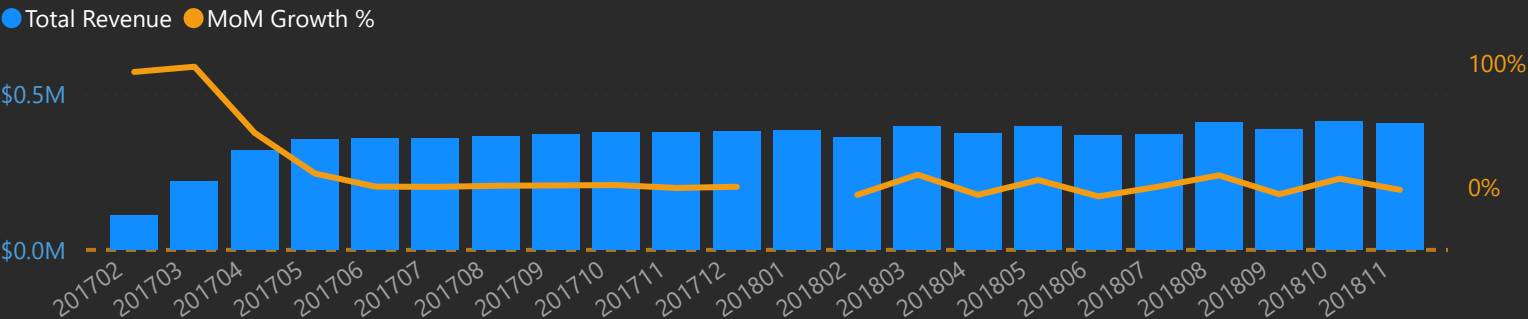
17.4%

Coupon Redemption Rate

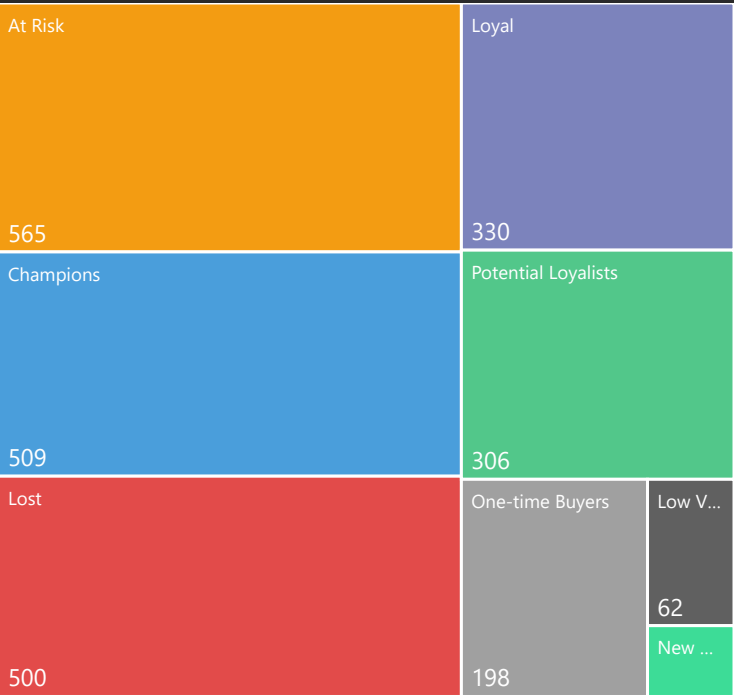
97.9%

Campaign Response Rate

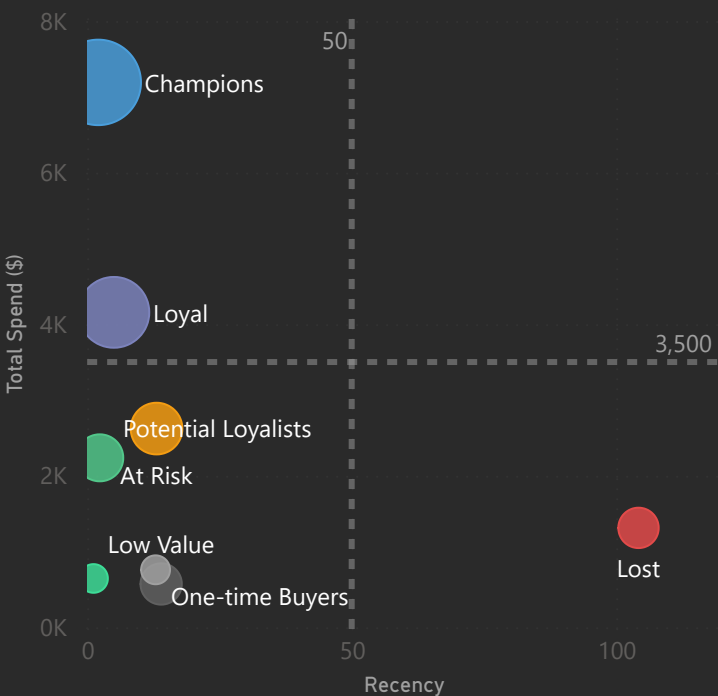
Monthly Revenue Trend



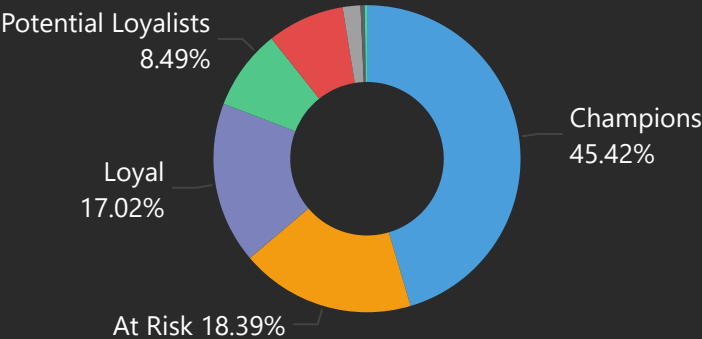
Household Distribution by Segment



Customer Landscape — Recency vs Spend



Revenue Contribution by Segment

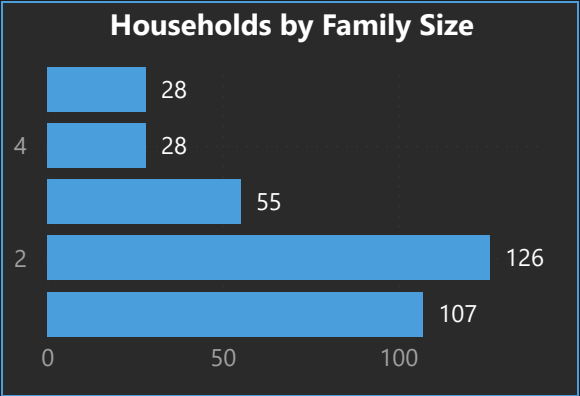
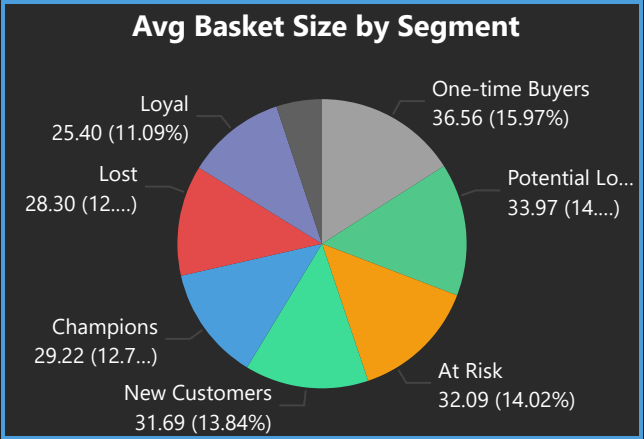
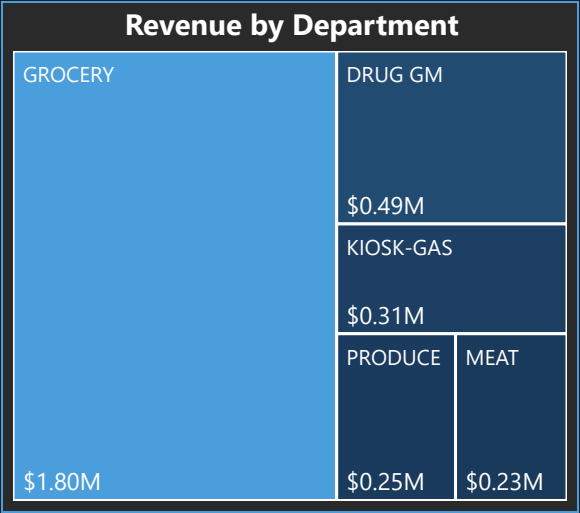
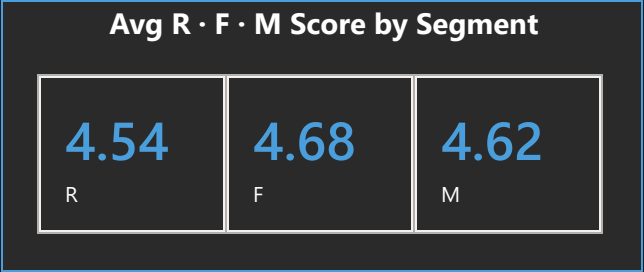
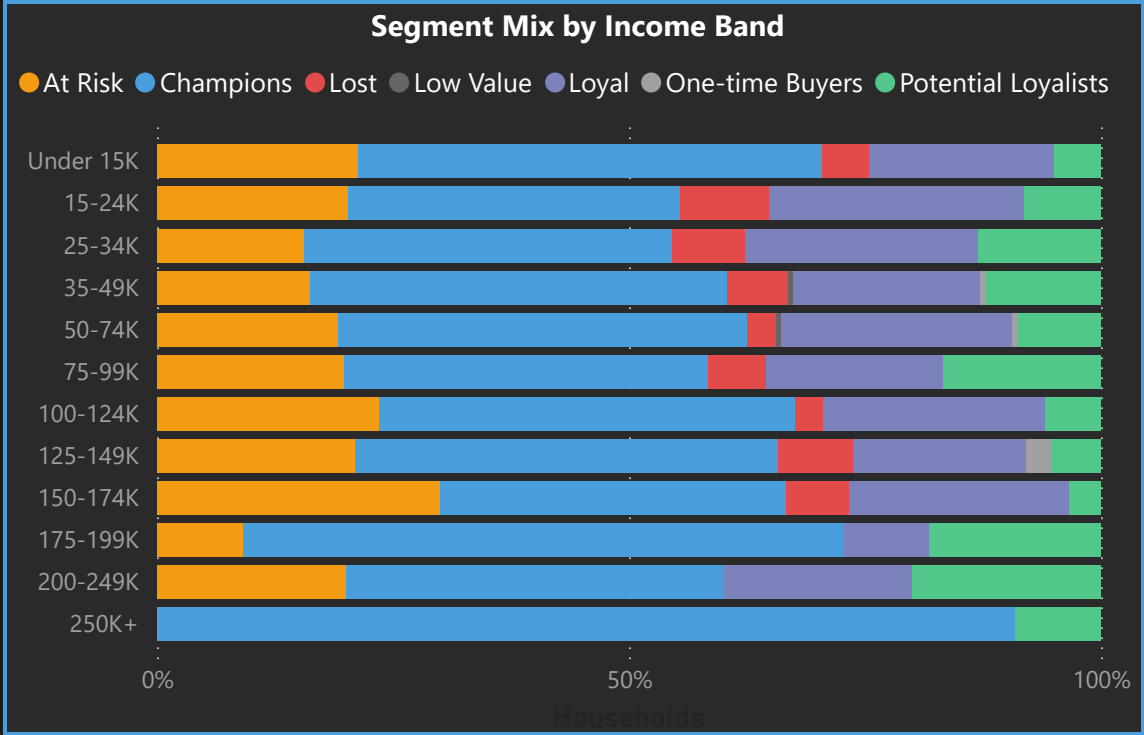
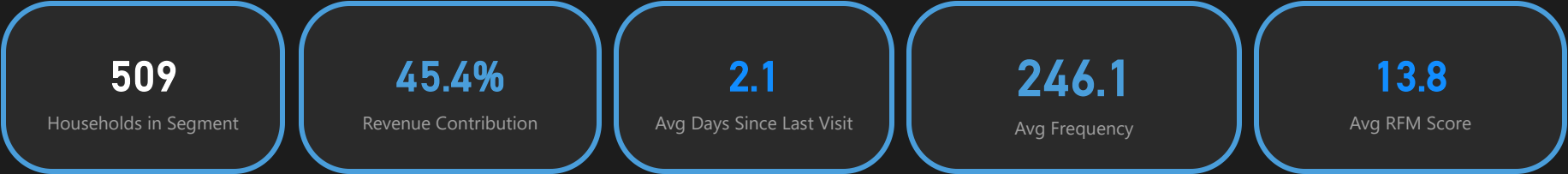
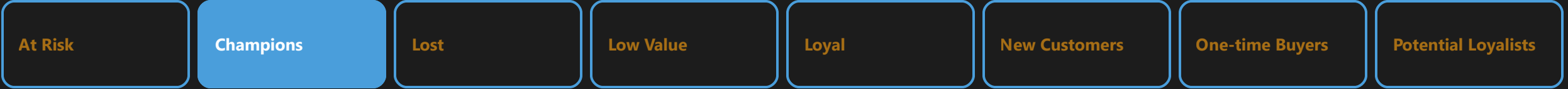


Key Finding: 42.6% of households show churn signals · Champions are 20.4% of customers but 45.4% of revenue, recency every 2 days drives their outsized value · Coupon spend concentrates on customers who buy anyway

Segment Deep-Dive — Who Are Our Customers?

Select a segment below to explore demographics and buying behaviour

Based on 801 households with known demographics (32% of total)



Key Finding: Higher income households skew Champion-heavy · 2-person households are the largest segment across all groups · National brands dominate at 70%+ regardless of segment

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At Risk

Champions

Lost

Low Value

Loyal

New Customers

One-time Buyers

Potential Loyalists

500

Households in Segment

8.1%

Revenue Contribution

104.2

Avg Days Since Last Visit

46.4

Avg Frequency

4.7

Avg RFM Score

Revenue by Department

GROCERY

DRUG GM

\$84.24K

MEAT

\$48.70K

PRODU...

MEAT-...

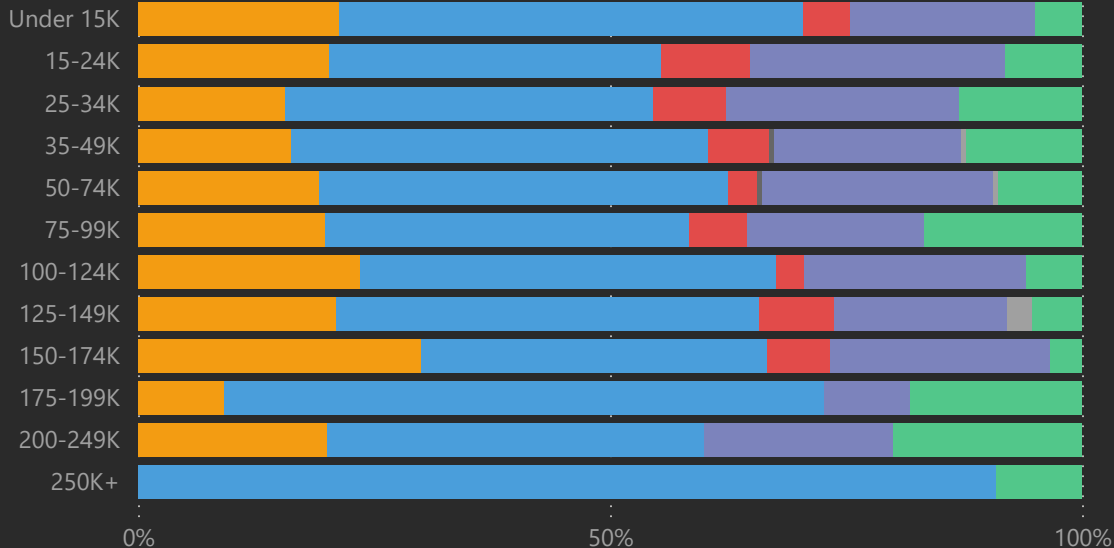
\$337.99K

\$40.62K

\$40.12K

Segment Mix by Income Band

At Risk Champions Lost Low Value Loyal One-time Buyers Potential Loyalists



Avg R · F · M Score by Segment

1.00

R

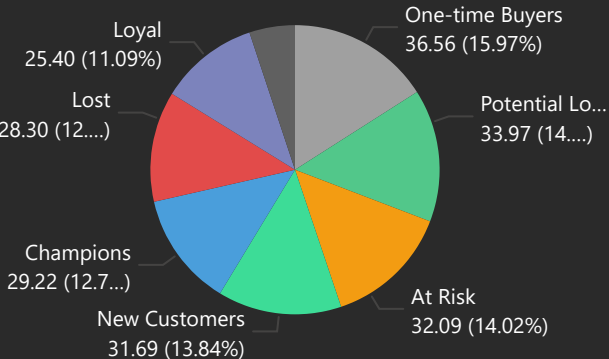
1.83

F

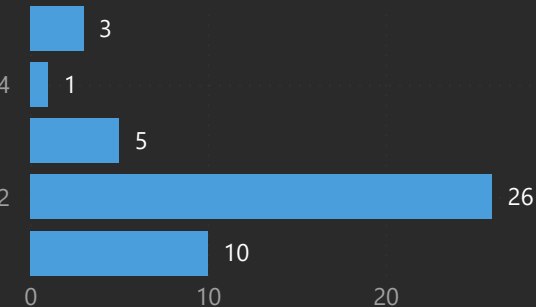
1.90

M

Avg Basket Size by Segment



Households by Family Size



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Potential Loyalists

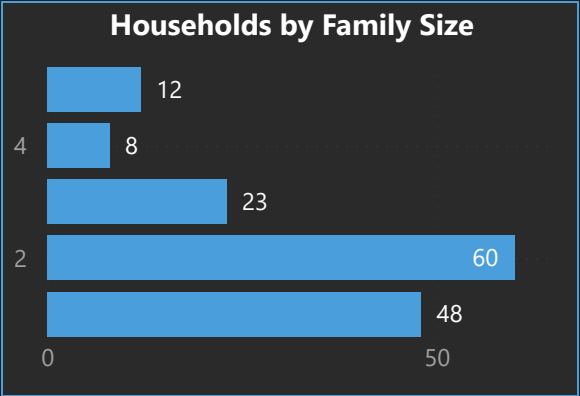
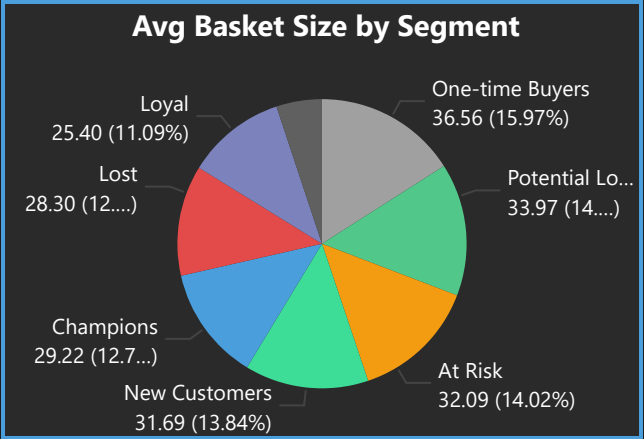
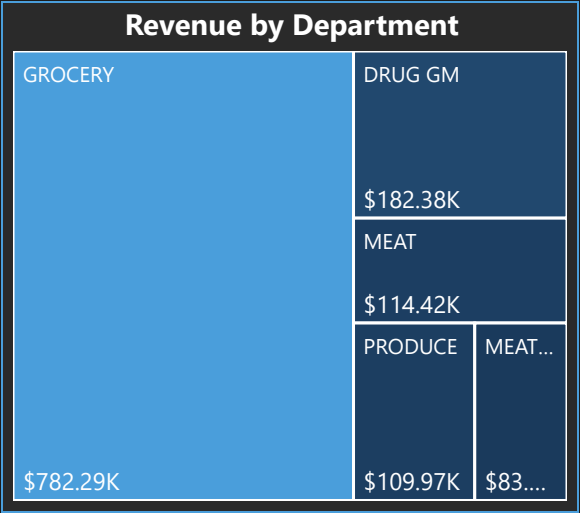
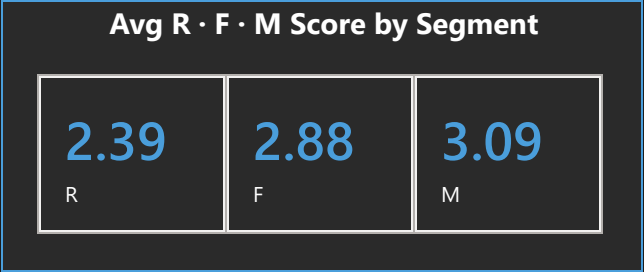
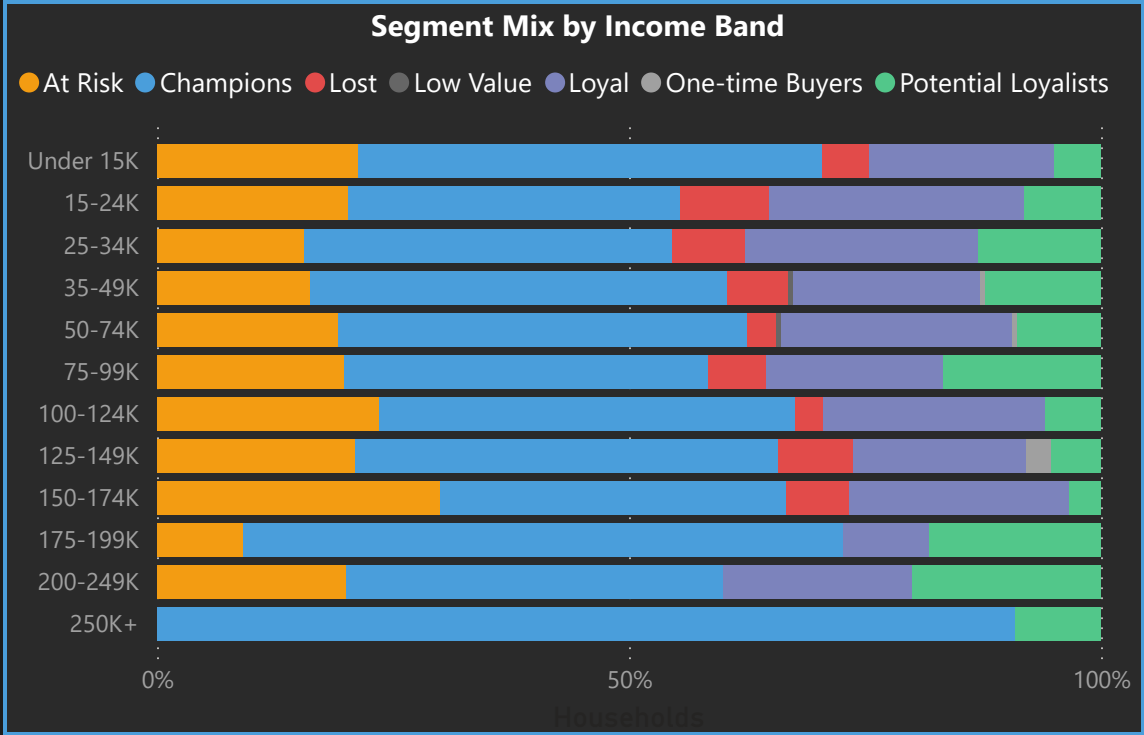
565
Households in Segment

18.4%
Revenue Contribution

13.1
Avg Days Since Last Visit

81.7
Avg Frequency

8.4
Avg RFM Score



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Based on 801 households with known demographics (32% of total)

At Risk

Champions

Lost

Low Value

Loyal

New Customers

One-time Buyers

Potential Loyalists

306

Households in Segment

8.5%

Revenue Contribution

2.4

Avg Days Since Last Visit

65.8

Avg Frequency

9.8

Avg RFM Score

Revenue by Department

GROCERY

DRUG GM

\$90.38K

PRODUCE

\$51.04K

MEAT

MEAT...

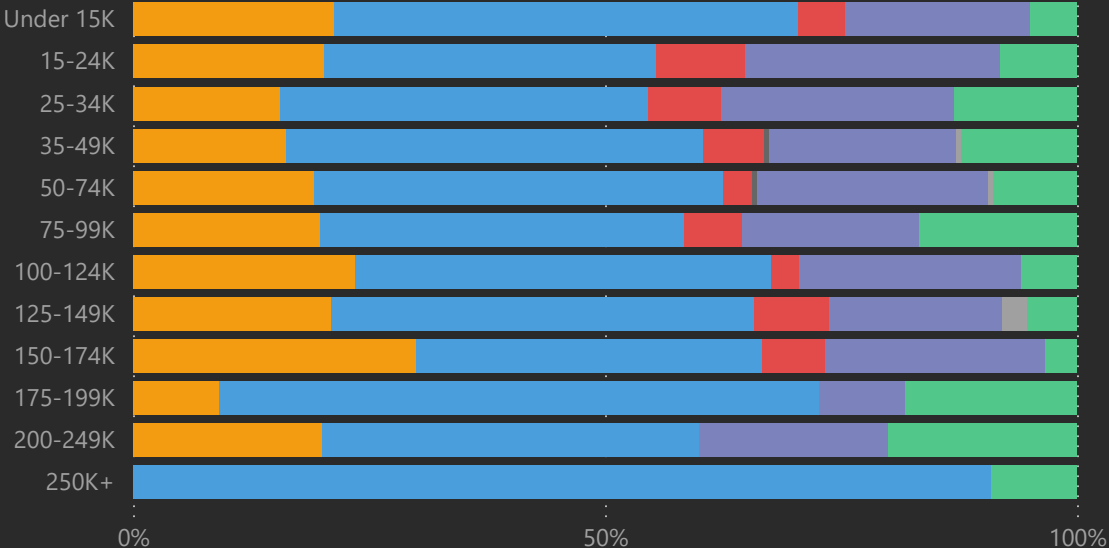
\$356.22K

\$49.55K

\$36....

Segment Mix by Income Band

At Risk Champions Lost Low Value Loyal One-time Buyers Potential Loyalists



Avg R · F · M Score by Segment

4.44

R

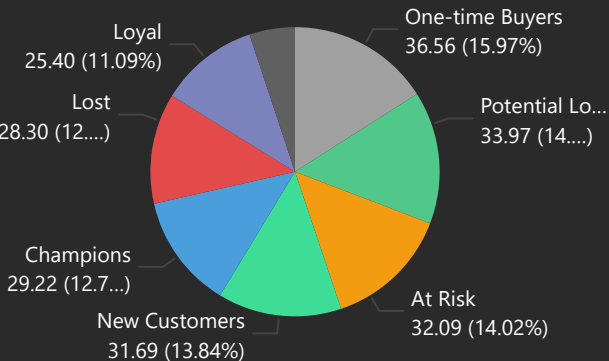
2.59

F

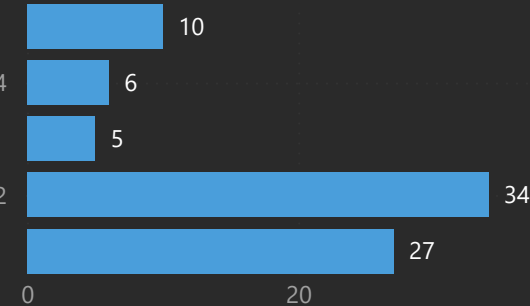
2.77

M

Avg Basket Size by Segment



Households by Family Size



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Segment Deep-Dive — Who Are Our Customers?

Select a segment below to explore demographics and buying behaviour

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At Risk

Champions

Lost

Low Value

Loyal

New Customers

One-time Buyers

Potential Loyalists

30

Households in Segment

0.2%

Revenue Contribution

1.2

Avg Days Since Last Visit

20.3

Avg Frequency

7.3

Avg RFM Score

Revenue by Department

GROCERY

DRUG GM

\$2.56K

PRODUCE

\$1.38K

MEAT

KIOSK-...

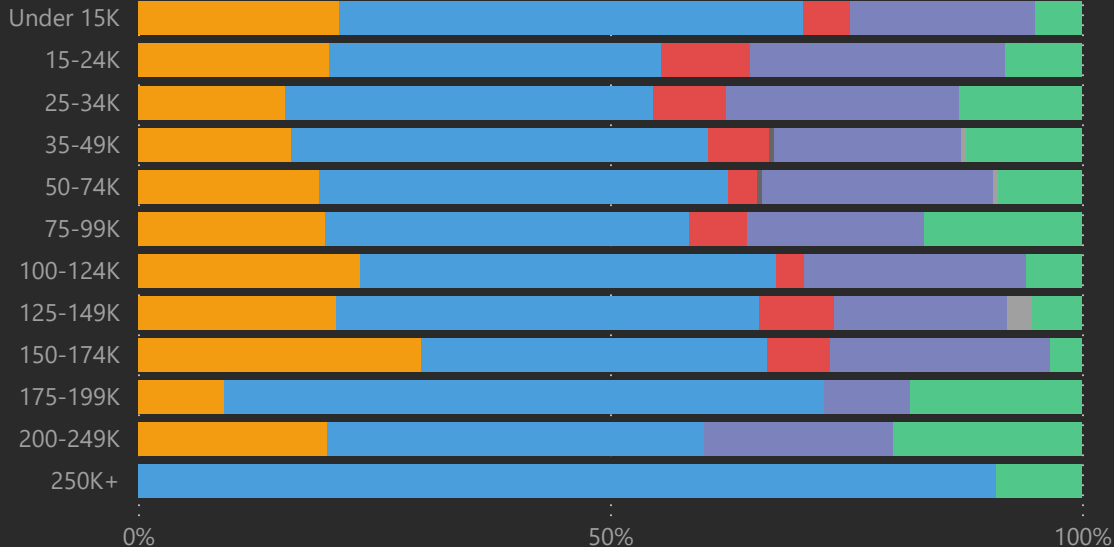
\$10.13K

\$1.17K

\$1.16K

Segment Mix by Income Band

At Risk Champions Lost Low Value Loyal One-time Buyers Potential Loyalists



Avg R · F · M Score by Segment

5.00

R

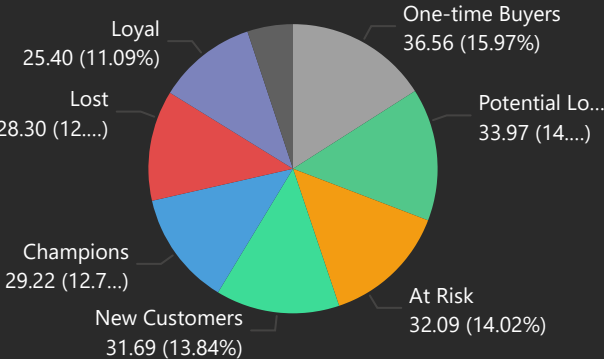
1.00

F

1.33

M

Avg Basket Size by Segment



Households by Family Size

Key Finding: Higher income households skew Champion-heavy · 2-person households are the largest segment across all groups · National brands dominate at 70%+ regardless of segment

Promotion Intelligence — Are We Wasting Spend?

Campaign response and coupon redemption analysis by customer segment

TypeA

TypeB

TypeC

30

Total Campaigns Run

1584

Targeted Households

916

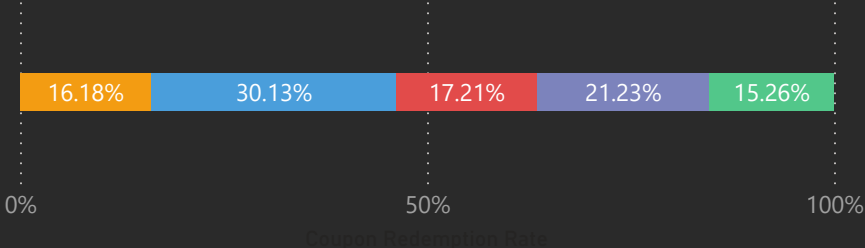
Never Targeted

17.4%

Coupon Redemption Rate

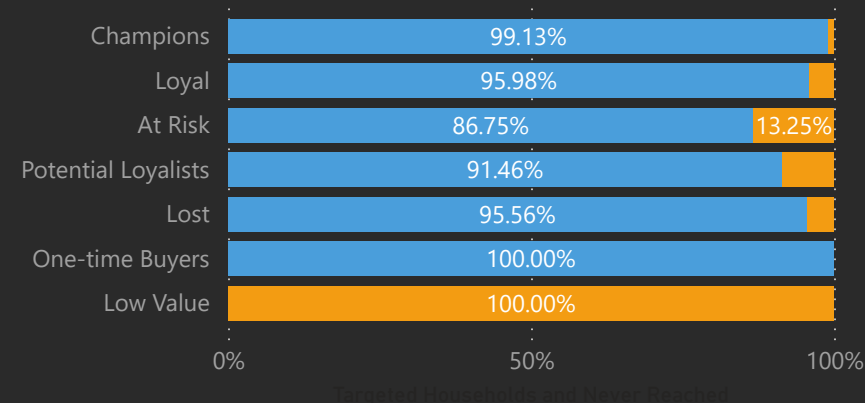
Share of Total Coupon Redemptions by Segment

At Risk Champions Lost Loyal Potential Loyalists

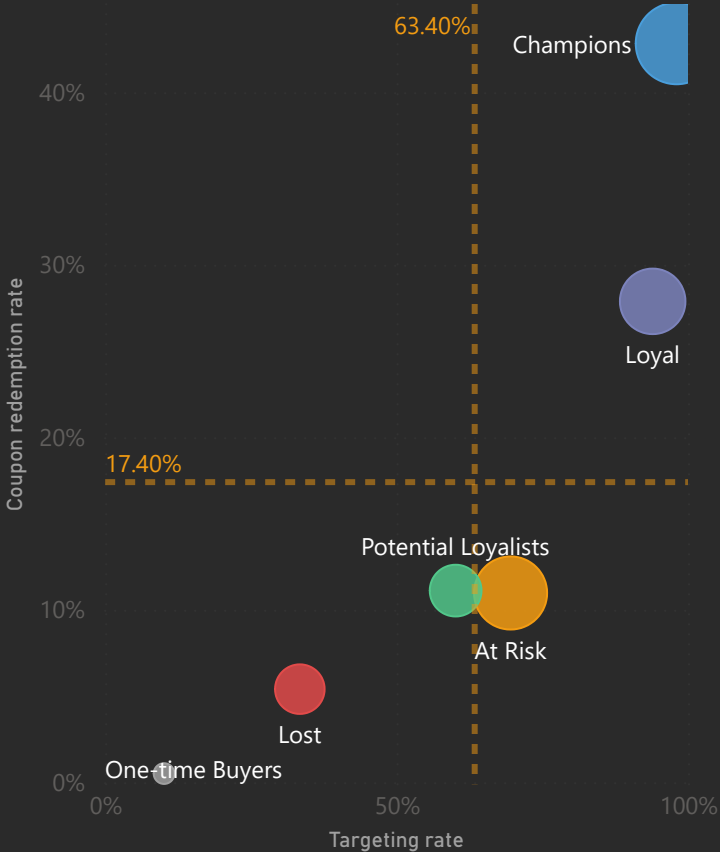


Targeted vs Untargeted Households by Segment

Targeted Households Never Reached



Targeting Efficiency vs Coupon Behaviour



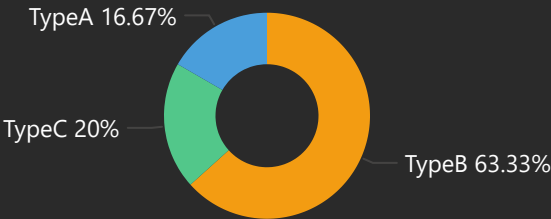
63.4%

Targeting Coverage

4.55

Avg Campaigns Received

Campaign Type Distribution



⚠ Response Rate Caveat 97.9% campaign response reflects baseline shopping frequency — not true campaign lift.
Avg campaign window = 47 days.
Median household shops every 9 days regardless of whether they received a campaign.

Key Finding: 916 households never targeted · Champions over-targeted at 99% yet absorb 60% of coupon value · Lost customers show lowest ROI at 5.4% redemption

Promotion Intelligence — Are We Wasting Spend?

Campaign response and coupon redemption analysis by customer segment

TypeA

TypeB

TypeC

5

Total Campaigns Run

1513

Targeted Households

987

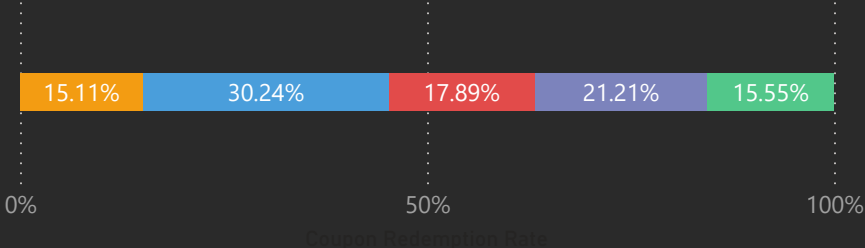
Never Targeted

15.3%

Coupon Redemption Rate

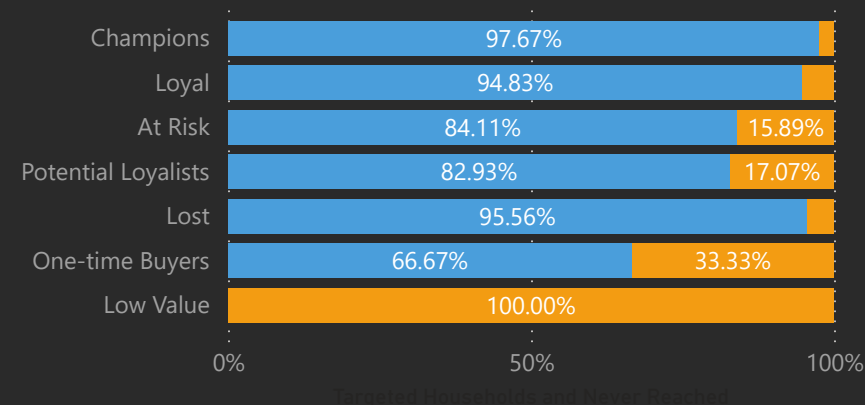
Share of Total Coupon Redemptions by Segment

At Risk Champions Lost Loyal Potential Loyalists

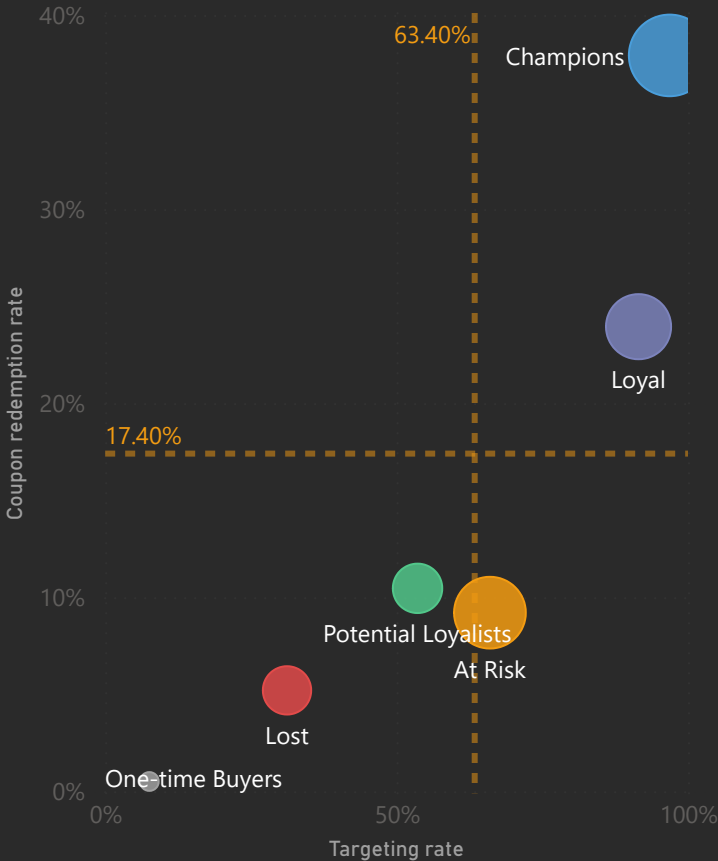


Targeted vs Untargeted Households by Segment

Targeted Households Never Reached



Targeting Efficiency vs Coupon Behaviour



60.5%

Targeting Coverage

2.63

Avg Campaigns Received

Campaign Type Distribution



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Promotion Intelligence — Are We Wasting Spend?

Campaign response and coupon redemption analysis by customer segment

TypeA

TypeB

TypeC

19

Total Campaigns Run

1023

Targeted Households

1477

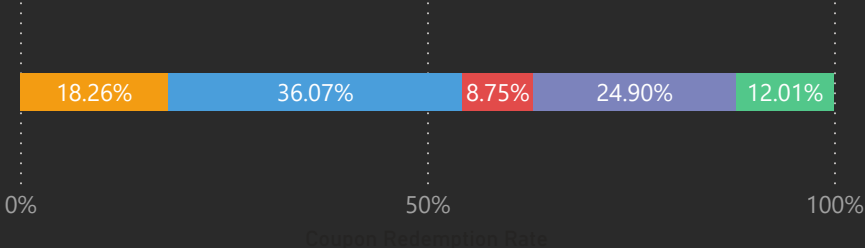
Never Targeted

5.6%

Coupon Redemption Rate

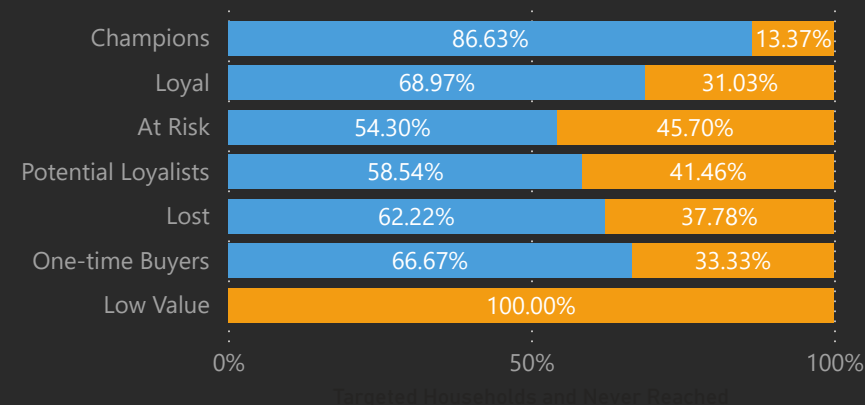
Share of Total Coupon Redemptions by Segment

At Risk Champions Lost Loyal Potential Loyalists

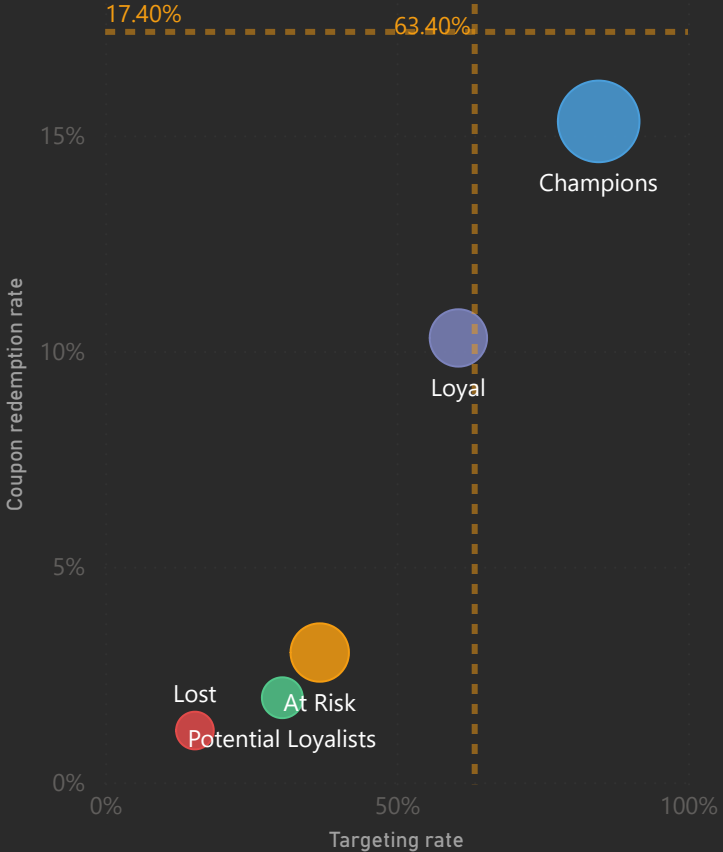


Targeted vs Untargeted Households by Segment

Targeted Households Never Reached



Targeting Efficiency vs Coupon Behaviour



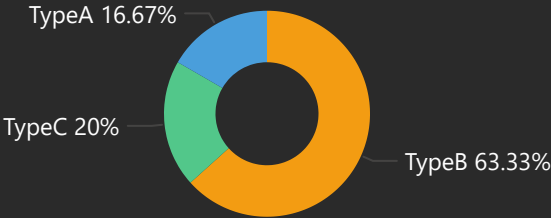
40.9%

Targeting Coverage

2.60

Avg Campaigns Received

Campaign Type Distribution

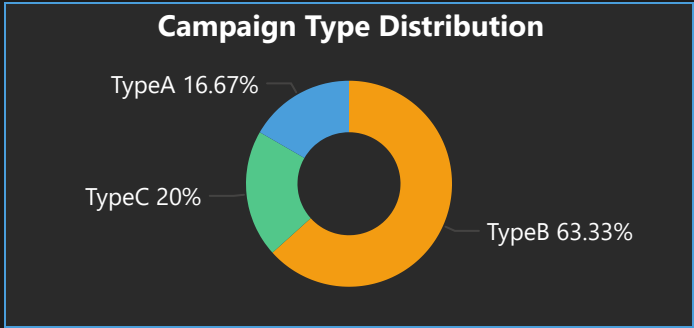
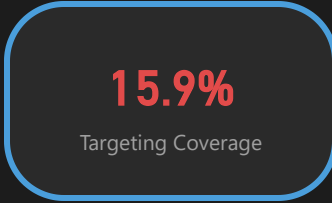
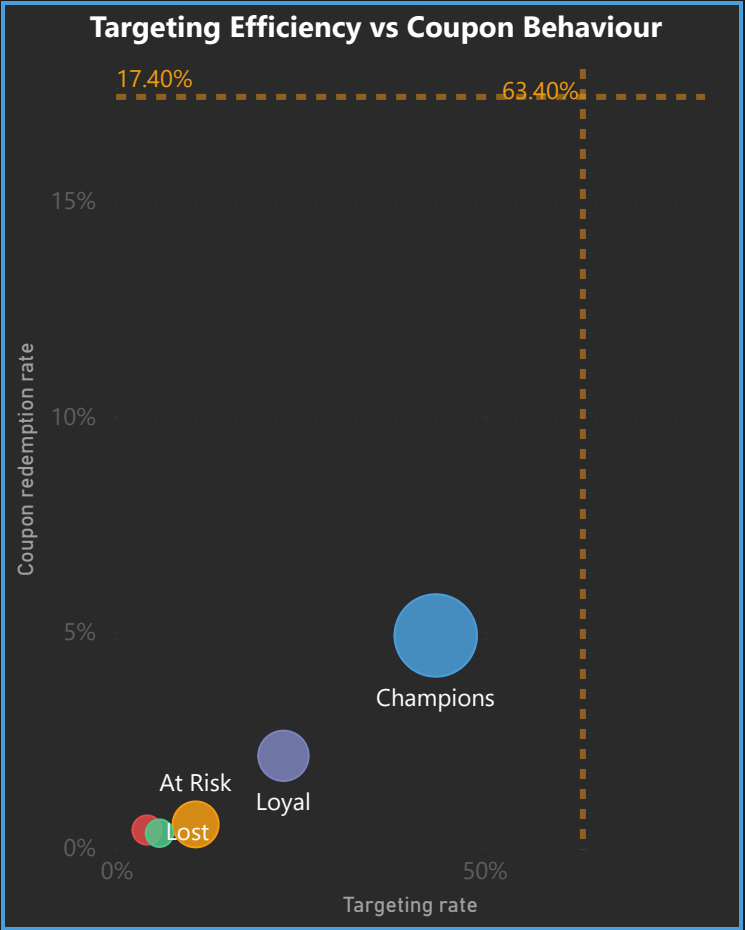
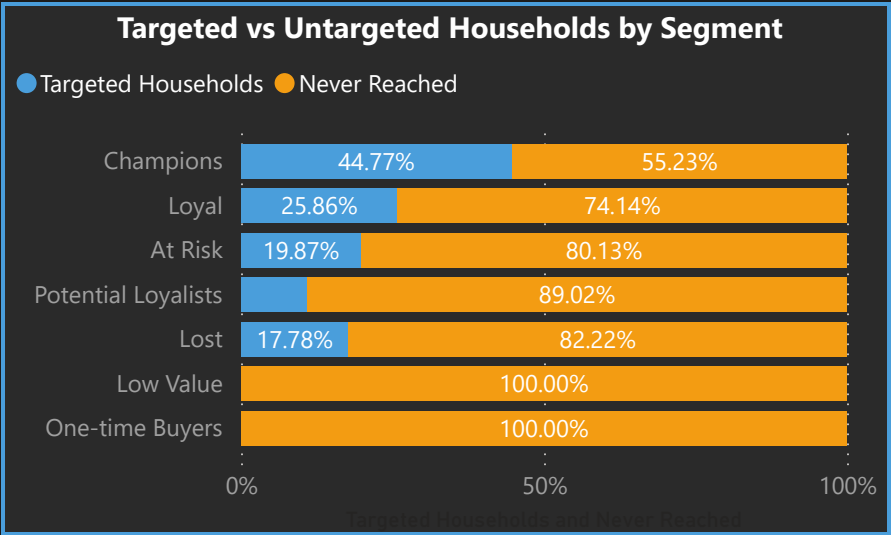
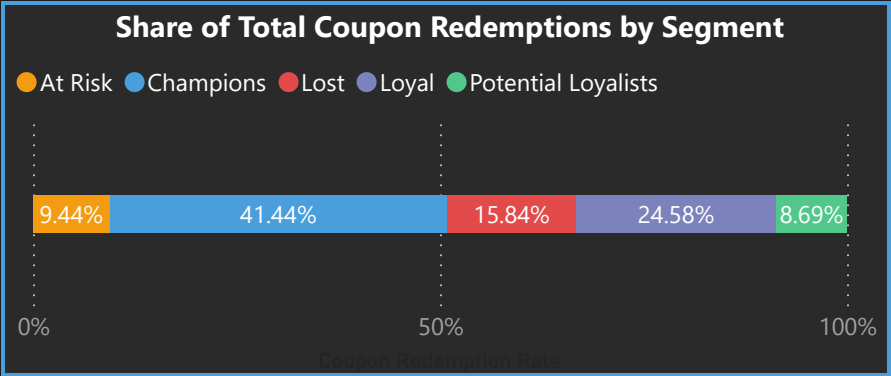
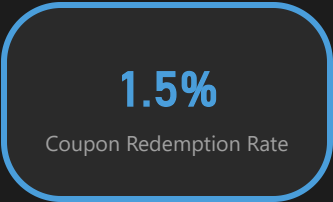
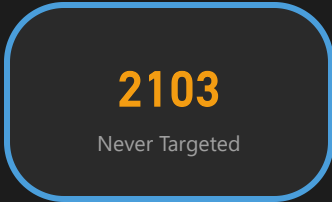
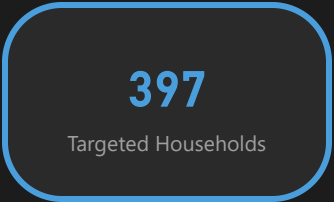


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